

ORF309: Probability and Stochastic Systems

Spring 2026: Midterm 1 Booklet

March 06, 09:35am – 10:25am

Instructor: Ioannis Akrotirianakis

Student Name (Print): _____

Student Email: _____

Honor Code: _____

Student Signature:

- Total number of problems: 3.
- Total number of points: 60.
- Please clearly write your solutions in the reserved space provided.

Problem 1. You roll two fair six-sided dice and tell me the product of the two numbers you get.

(a) [10 points] What is a good sample space for this random experiment? How to define the corresponding probability measure on it?

(b) [10 points] What is the expectation of the number you tell me?

Solution of Problem 1.

Problem 2. Researchers have observed that during the winter, students' willingness to attend morning lectures depends on weather conditions. On a day without snowfall, each student attends the lecture independently with probability $p_1 = 0.7$. On a snowy day, each student attends independently with probability $p_2 = 0.1$. On any given winter day, snow occurs with probability $q = 0.2$.

Assume there are 100 students enrolled in the class, and that attendance decisions are mutually independent across students, conditional on the weather.

(a) [10 points] What is the expected number of students attending the lecture on a random winter day?

(b) [10 points] Suppose that on a particular observed day, exactly 30 students attended the lecture. Given this information, what is the probability that it snowed on that day?

Solution of Problem 2.

Solution of Problem 2, continued.

Problem 3. Alice and Bob play a game involving a fair six-sided die and a fair coin. In each round, Alice rolls the die and Bob flips the coin. Alice earns \$3 each time she rolls a 6, and Bob earns \$1 each time he flips a head. All outcomes are assumed to be mutually independent across players and across rounds.

(a) [10 points] If the game lasts for 10 rounds, compare the variances of their total earnings.

(b) [10 points] If the game lasts until Alice earns \$3, what is the variance of Bob's total earning?

You may use the following fact from Calculus without justification:

$$\sum_{n=1}^{\infty} n^2 p (1-p)^{n-1} = \frac{2-p}{p^2}, \quad \forall p \in (0, 1).$$

Solution of Problem 3.

Solution of Problem 3, continued.